Module 3 solution

```
source("R/prepare_everything.R")
```

```
[1] "Metadata imported!"
[1] "Chronos scores imported!"
```

```
[1] "Expression data imported!"
[1] "Mutation data imported!"
```

Module 3: Genetic buffering

For specific genes, it has been explored that they can fulfill the function of another gene. This is specifically relevant for highly essential gene functions, which can be often fulfilled by multiple genes.

The buffering potential of a gene on another can be quantified if the expression values of one gene correlate of one genes Chronos scores of another gene.

3.1:

Lets compute the correlation between the expression values of `CDS1` and the Chronos scores of `CDS2` using `cor()`.

```
intersect_cell_lines <- intersect(rownames(expression), rownames(chronos))

cor(expression[intersect_cell_lines,"CDS1"], chronos[intersect_cell_lines,"CDS2"])
```

```
[1] 0.6461067
```

3.2

Compute the correlation for all genes from the expression set with the Chronos scores of `CDS2`. Is CDS1 an extreme value or rather average?

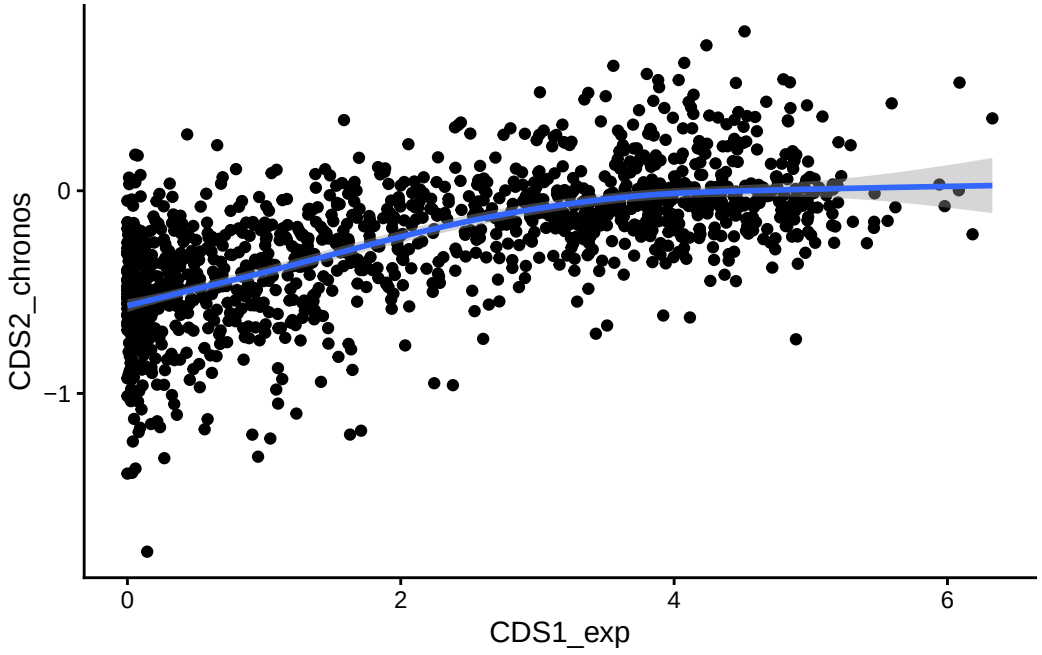
3.3

Plot the CDS1 expression on the X axis and the CDS2 Chronos scores on Y. Create a scatter plot (`geom_point()`). Also add a fit to visualize the trend `geom_smooth()`.

```
plot_data <- data.table(cell_line = intersect_cell_lines,
                        CDS1_exp = expression[intersect_cell_lines,"CDS1"],
                        CDS2_chronos = chronos[intersect_cell_lines,"CDS2"])

ggplot(plot_data, aes(CDS1_exp, CDS2_chronos)) +
  geom_point() +
  geom_smooth()
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



Challenge:

Try the same plot, but color the points by their `OncotreeLineage`. You should move the `color` aesthetics to `geom_point()`, otherwise `geom_smooth()` tries to compute many fits for very few datapoints, causing a ton of warnings and error messages.

```
plot_data <- models[intersect_cell_lines, on = "ModelID",
                    .(ModelID,
                      CDS1_exp = expression[intersect_cell_lines,"CDS1"],
                      CDS2_chronos = chronos[intersect_cell_lines,"CDS2"],
                      OncotreeLineage)]

ggplot(plot_data, aes(CDS1_exp, CDS2_chronos, color = OncotreeLineage)) +
  geom_point() +
  geom_smooth(na.rm = TRUE) +
  theme(legend.position = "bottom")
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: span too small.  fewer data values than degrees of freedom.

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: pseudoinverse used at 1.9907

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: neighborhood radius 0.81218

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: reciprocal condition number 0

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: There are other near singularities as well. 0.41464

Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
else if (is.data.frame(newdata))
as.matrix(model.frame(delete.response(terms(object)), : span too small.  fewer
data values than degrees of freedom.

Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
else if (is.data.frame(newdata))
as.matrix(model.frame(delete.response(terms(object)), : pseudoinverse used at
1.9907
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Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
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Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: at 1.9164

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: radius 1.8825e-05

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: all data on boundary of neighborhood. make span bigger

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: pseudoinverse used at 1.9164

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: neighborhood radius 0.0043388

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: reciprocal condition number 1

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: at 2.7929
```

```
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: radius 1.8825e-05

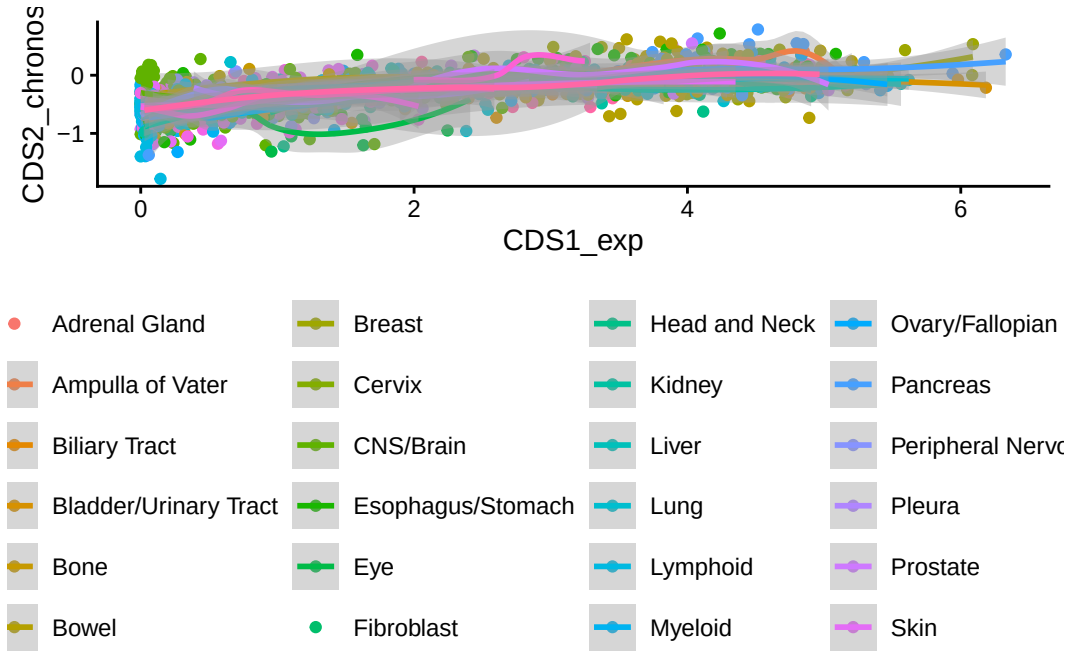
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: all data on boundary of neighborhood. make span bigger

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: There are other near singularities as well. 1.8825e-05

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: zero-width neighborhood. make span bigger
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: zero-width neighborhood. make span bigger

Warning: Failed to fit group 29.
Caused by error in `predLoess()`:
! NA/NaN/Inf in foreign function call (arg 5)

Warning in max(ids, na.rm = TRUE): no non-missing arguments to max; returning
-Inf
```



```
plot_data <- models[intersect_cell_lines, on = "ModelID",
                    .(ModelID,
                      CDS1_exp = expression[intersect_cell_lines,"CDS1"],
                      CDS2_chronos = chronos[intersect_cell_lines,"CDS2"],
                      OncotreeLineage)]

ggplot(plot_data, aes(CDS1_exp, CDS2_chronos)) +
  geom_point(aes(color = OncotreeLineage)) +
  geom_smooth(na.rm = TRUE) +
  theme(legend.position = "bottom")
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

